An Affordable Recipe for Training Diffusion Transformers

Dog-Breed Text-to-Image Diffusion, from Pretraining to Fine-Tuning, on 4 Consumer GPUs

Submission to the NeurIPS 2026 Call for Educational Resources

Training diffusion transformers is usually presented as something that requires industrial-scale compute. This resource teaches the opposite lesson through a real, runnable pipeline: the recipe below was arrived at over two months of empirically comparing image tokenizers, latent-space parameterizations, and training hyperparameters, converging on a configuration that pretrains a working text-to-image Diffusion Transformer (DiT) to sub-10 FID in 2.5 hours, and fine-tunes it in 20 more minutes, on hardware any student can rent by the hour. Students run this recipe themselves and evaluate their own results — no pretrained weights are distributed with the resource, so every checkpoint you get is your own.

~2h50m total training time on 4×RTX 3090 (24GB) → sub-10 FID
about $3–$5 of rented compute for the whole pipeline

2.5h pretrain (Exp. 1) + 20min fine-tune (Exp. 2). GPU pricing: Vast.ai $0.25/GPU/hr, RunPod $0.46/GPU/hr (Aug 2026 quoted rates).

Learning objectives

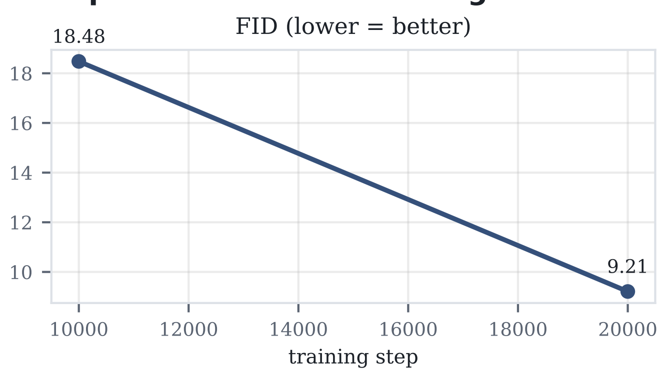
- Explain latent-space diffusion and why it makes transformer-based image generation tractable.
- Derive and implement the flow-matching training objective (straight-line paths, velocity prediction).
- Contrast in-context (token-based) conditioning against AdaLN-Zero conditioning in DiT architectures.
- Apply REPA-style auxiliary losses that align internal DiT features with a pretrained vision encoder.
- See how tokenizer and recipe choices, not scale, determine whether DiT training is affordable on consumer GPUs.
- Build a full data pipeline: downloading raw image/caption shards and precomputing a training-ready dataset.
- Diagnose what a fine-tuning FID curve is really measuring using a real evaluation metric, not training loss alone.
- Score generations with learned human-preference models (PickScore, HPSv2) and reconcile them against FID/IS.

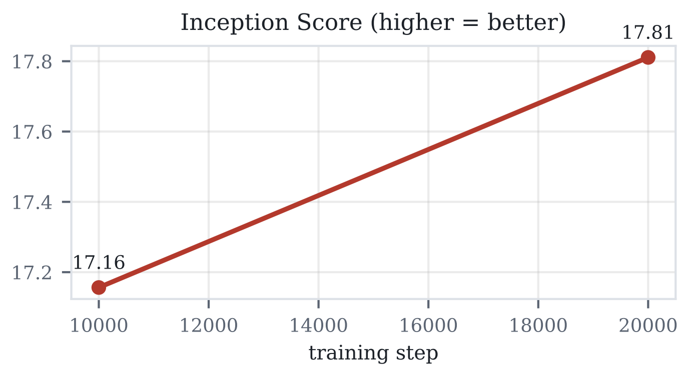
What students actually do

The notebook is organized into 13 sections: background theory tied to the real training code, each model component explained and run individually (VAE tokenizer, text encoder, REPA target, the DiT backbone itself), the full data-preparation pipeline, the pretraining run, the fine-tuning run, inference/evaluation against FID and Inception Score, and a final section scoring the same generations with learned human-preference models. Every code cell is real, executable code from the actual research codebase, not a simplified reimplementation.

Results below are from my own runs, shown as a concrete example of the phenomena students are asked to reproduce and diagnose themselves in Sections 10 and 12 of the notebook.

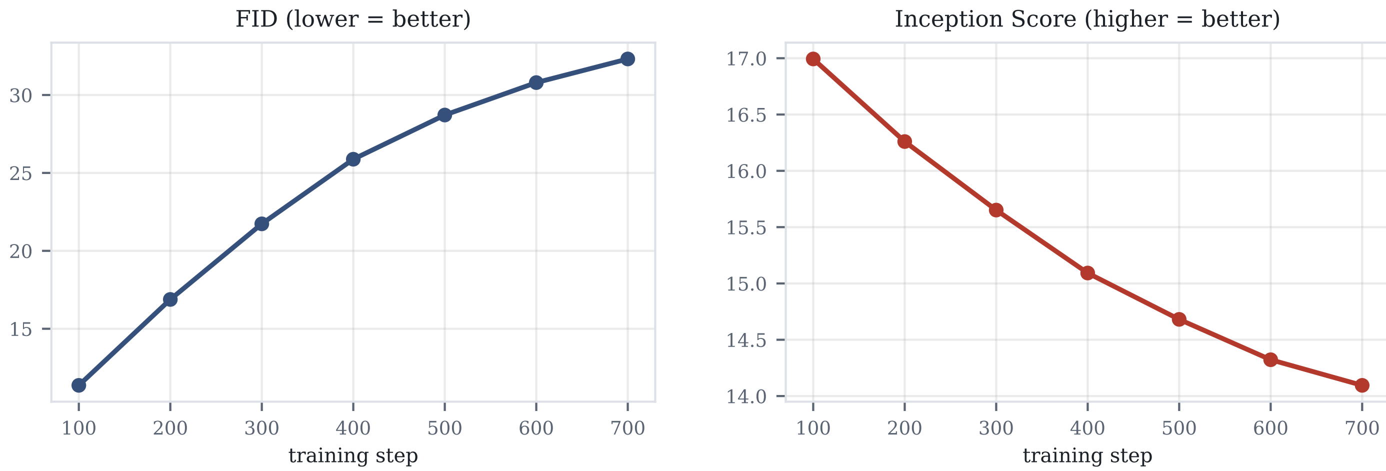
Experiment 1 — Pretraining from scratch





vavae-repa-recaptioned-latents-invae-EMA0.9995-200Epochs — FID improves (18.5→9.2) and Inception Score improves (17.2→17.8) as pretraining progresses: healthy convergence, the expected pattern.

Experiment 2 — Fine-tuning (SFT) on a smaller dataset



invae-sft-dogs-synthetic-2k-EMA0.995-100ep-5 — FID rises monotonically (11.4→32.3) and Inception Score falls monotonically (17.0→14.1) across the run — see below for why this is not simply "the model got worse," on a fine-tuning set of only 2,000 images.

Experiment 3 — Learned human-preference scoring (PickScore, HPSv2)

	HPSv2 mean score	PickScore: preferred over other model
Pretrained	0.2409	2.4% of 500 pairs
SFT	0.2719	97.6% of 500 pairs

500 held-out captions, one image per checkpoint per caption, same seed. Both reward models — trained on real human preference judgments, independent of FID/IS — agree the SFT checkpoint is preferred.

Why these results are the resource's central lesson

Three real, independent results, not one — and they don't all point the same way, which is exactly the point. Pretraining shows healthy, monotonic improvement in FID/IS. Fine-tuning the same architecture on a much smaller set shows FID/IS moving in the opposite direction for the whole run — yet the learned preference models say the final SFT checkpoint is still clearly preferred over never fine-tuning at all.

Why does FID rise rather than fall? Most likely a metric artifact, not overfitting: both evaluation stages reuse the identical FID reference statistics, built once from the 26,000-image pretraining photo set — including for the SFT model, whose actual training data (2,000 curated synthetic images) is a deliberately different distribution. FID measures distance to whichever reference it's given, so specializing toward a new distribution raises FID against the old one regardless of quality. The checkpoint scored in Experiment 3 — ep-0000100, the final and single worst-FID checkpoint above — is still the one PickScore/HPSv2 prefer 97.6% of the time, hard to square with "the images got worse" and easy to square with a mismatched reference. Neither metric alone tells the complete story here.

Students do not see these results before running the notebook — Sections 10 and 12 have them reproduce and reconcile all three themselves, on their own trained checkpoints.

Resource access

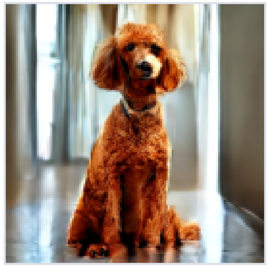
Code, configs, and the full tutorial notebook (dog_t2i_diffusion_tutorial.ipynb):
https://github.com/Reyhaneesmailizadeh/diffusion-bench

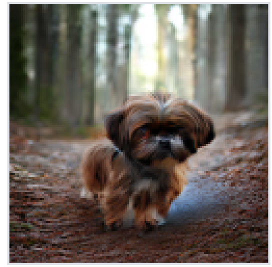
No pretrained weights are distributed with this resource — students download the raw datasets and train both stages themselves, on GPU compute of their own choosing (the notebook documents the exact commands used at the scale run for this submission).

Sample generations from the SFT checkpoint

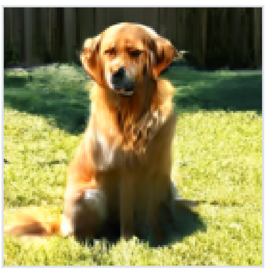
invae-sft-dogs-synthetic-2k-EMA0.995-100ep-5 / ep-0000100 — the same checkpoint scored in Experiment 3. One generated image per breed shown, from the 500-caption reward-model evaluation set (Section 12).

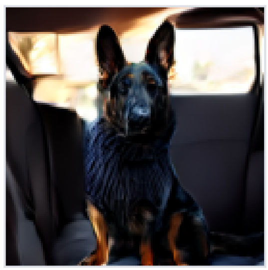

Yorkshire Terrier

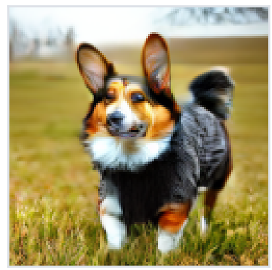

Standard Poodle


Shih Tzu

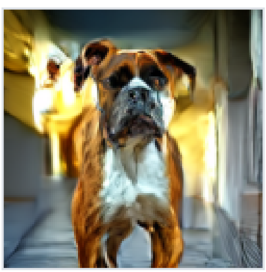

Samoyed

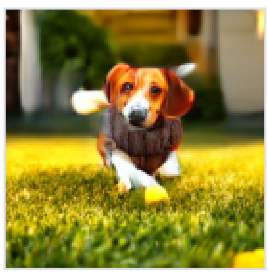

Golden Retriever

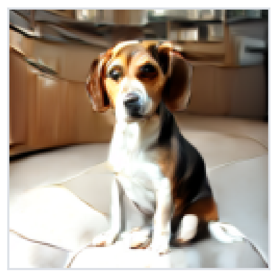

German Shepherd

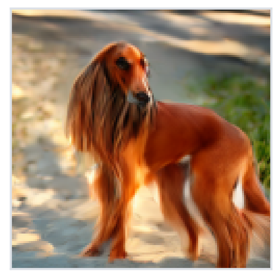

Corgi

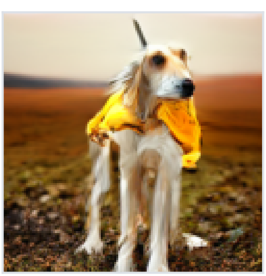

Chow


Boxer


Beagle


Beagle


Afghan Hound


Afghan Hound